# Housing Affordability Dynamics: Price-to-Rent Ratios Across U.S. Metro Areas, 2020–August 2026

Thao Nguyen
DATA 400 – Individual Mini-Project
Disciplinary Track: Finance / Economics

Fall 2026

**Abstract**

This project examines how home values and rents evolved across the 100 largest U.S. metropolitan areas from January 2020 through August 2026. Using Zillow's Home Value Index (ZHVI) and Observed Rent Index (ZORI), the analysis constructs a price-to-rent (PTR) ratio, compares cumulative home-value and rent growth, ranks metros by the difference between these growth rates, explores lagged associations between rent and home-value growth, and groups metros using K-means clustering. Across the selected metros, home values rose by an average of 48.1% while rents rose by 42.3%, a mean growth gap of 5.7 percentage points. The median PTR ratio peaked at approximately 18.1 in late 2022 and eased to about 17.2 by August 2026, remaining above its January 2020 level of about 16.7. McAllen, Texas, had the largest positive growth gap, while San Jose, California, had the largest increase in PTR. Rent growth was strongly associated with contemporaneous home-value growth, but its relationship with home-value growth twelve months later was weaker; the corresponding regression had an $R^2$ of approximately 0.17. These results describe differences in market trajectories, not definitive measures of individual affordability or causal relationships.

## 1 Introduction and Research Question

Housing is a major household expense and an important component of household financial decisions. Choosing whether to rent or buy depends on more than the current monthly payment: households may also consider home prices, expected rent changes, financing costs, income, maintenance, taxes, and how long they expect to remain in a location. Comparing the evolution of home values and rents therefore offers one useful way to describe changing housing-market conditions.

The COVID-19 period and its aftermath brought substantial changes to housing markets across the United States. However, home values and rents did not necessarily rise at the same pace in every metropolitan area. If home values increase faster than rents, the price-to-rent ratio tends to rise, meaning that a home purchase price is higher relative to the annual rent represented by the index. If rents rise faster than home values, the ratio tends to fall. Neither direction alone proves that a market is overvalued or that renting is always financially preferable, but the ratio provides a consistent descriptive measure for comparing places and time periods.

The primary research question is:

> *How did the relationship between home values and rents change across the 100 largest U.S. metropolitan areas from January 2020 through August 2026, and which metros experienced the largest differences between home-price growth and rent growth?*

The project also considers four related questions: (1) How did the median price-to-rent ratio change over time? (2) Which metros experienced the largest positive or negative difference between home-value growth and rent growth? (3) Can metros be grouped according to their home-value and rent growth patterns? and (4) Is rent growth associated with subsequent home-value growth?

# 2 Disciplinary Motivation

This project combines finance, economics, and data analytics. From a real-estate-finance perspective, the price-to-rent ratio compares an asset's purchase value with the rental flow associated with a similar housing market. It is a useful descriptive benchmark, although it does not account for mortgage rates, down payments, property taxes, insurance, maintenance, transaction costs, or expected appreciation. From an economics perspective, differences between home-price and rent growth may reflect variation in local demand, housing supply, population movements, income, construction, and other market conditions. This project does not directly estimate the effects of those factors; instead, it documents observed patterns that could motivate further investigation.

The data-analytics component involves reshaping wide datasets into a monthly panel, cleaning and merging records, constructing interpretable features, visualizing time-series and cross-sectional variation, estimating an exploratory regression, and applying an unsupervised clustering method. The central goal is descriptive and comparative rather than causal. In particular, the PTR ratio is not treated as a stand-alone test of a housing bubble, and associations between rent and price growth are not interpreted as evidence that one causes the other.

# 3 Data and Sample

## 3.1 Data Sources

The analysis uses two publicly available datasets from Zillow Research's housing data portal [1]:

1. **Zillow Home Value Index (ZHVI):** a monthly index of typical home values. The downloaded series used for this project represents the 33rd–67th percentile home-value tier and is smoothed and seasonally adjusted.

2. **Zillow Observed Rent Index (ZORI):** a monthly index of typical observed rents. The downloaded series used here is smoothed and covers single-family and condominium units.

Both files are provided at the metropolitan-area level and originally contain historical observations extending earlier than the study period. The analysis filters the observations to January 2020 through August 2026. The unit of observation is a metropolitan statistical area by month. The key fields are metro name, Zillow's metro SizeRank, month, ZHVI, and ZORI. The index values are measures of typical market conditions; they do not track the exact purchase price or rent paid by a particular household.

## 3.2 Sample Selection and Preparation

The starting sample is the 100 largest metro areas according to Zillow's SizeRank. Because the ZORI series does not have equally complete historical coverage for every metro, the project evaluates coverage and retains metros with sufficient overlapping observations. The presentation's coverage rule is at least 76 monthly observations, corresponding to 95% of the 80-month study window. After reshaping the source files from wide to long format, standardizing date and metro fields, filtering the period, and merging on metro and month, the resulting panel contains approximately 8,000 metro-month observations. The exact number of usable rows can depend on missing-value handling and should be confirmed by rerunning the notebook.

The main preparation steps are:

1. Read the ZHVI and ZORI CSV files.

2. Convert the monthly date columns into a long-format date variable and value column.

3. Restrict dates to January 2020–August 2026 and metros to the selected top-100 sample.

4. Check missingness and overlapping coverage, then merge the datasets by metro and month.

5. Construct the ratio and growth measures described below.

# 4 Methods

## 4.1 Price-to-Rent Ratio

For metro $i$ in month $t$, let $P_{it}$ denote ZHVI and $R_{it}$ denote ZORI. Annualized rent is defined as

$$A_{it} = 12R_{it}. \tag{1}$$

The price-to-rent ratio is then

$$PTR_{it} = \frac{P_{it}}{12R_{it}}. \tag{2}$$

This ratio compares the typical home-value index with twelve months of typical rent. It is a relative-price indicator, not a complete rent-versus-buy financial calculation.

## 4.2 Cumulative Growth and Divergence

For each metro, cumulative home-value and rent growth compare the first and last available observations in the study window:

$$G_i^P = 100\left(\frac{P_{i,\text{end}}}{P_{i,\text{start}}} - 1\right), \tag{3}$$

$$G_i^R = 100\left(\frac{R_{i,\text{end}}}{R_{i,\text{start}}} - 1\right). \tag{4}$$

The growth divergence is defined as

$$D_i = G_i^P - G_i^R. \tag{5}$$

A positive $D_i$ means home values increased more than rents over the period; a negative value means rents increased more than home values. The change in the price-to-rent ratio is calculated as

$$\Delta PTR_i = PTR_{i,\text{end}} - PTR_{i,\text{start}}. \tag{6}$$

For the overall trend comparison, each series is also indexed to 100 in January 2020:

$$I_{it} = 100\frac{X_{it}}{X_{i,\text{Jan2020}}}, \tag{7}$$

where $X$ is either ZHVI or ZORI. Year-over-year growth is calculated as

$$YoY_{it} = 100 \left( \frac{X_{it}}{X_{i,t-12}} - 1 \right). \tag{8}$$

### 4.3 Lagged Association and Regression

To explore whether rent growth is associated with later home-value growth, the analysis calculates correlations between rent year-over-year growth at time $t$ and home-value year-over-year growth at $t + k$, for lags $k$ from 0 to 24 months. It also estimates a simple twelve-month-ahead regression:

$$G_{i,t+12}^{P,YoY} = \beta_0 + \beta_1 G_{it}^{R,YoY} + \varepsilon_{it}. \tag{9}$$

The regression is exploratory. It does not control for mortgage rates, income, housing supply, employment, migration, or metro-specific characteristics. The coefficient therefore describes an association in this sample rather than a causal effect or a complete forecasting model.

### 4.4 K-Means Clustering

The project applies K-means clustering to standardized metro-level cumulative home-value growth and rent growth, using $k = 3$. Standardization places the two growth measures on comparable scales before the algorithm groups metros according to their joint patterns. The resulting clusters are summarized as fast-growing, middle, and slow-growing groups based on their average growth rates. These labels are descriptive shorthand, not official categories or claims about the quality of a market. Results may change with a different choice of $k$, sample, feature set, or scaling method.

## 5 Findings

### 5.1 Home Values Increased More Than Rents on Average

Across the 100 selected metros, mean cumulative home-value growth was 48.1%, compared with 42.3% for rents. The difference between these two mean growth rates was approximately 5.7 percentage points. At the metro level, median home-value growth was 49.5% and median rent growth was 43.0%, yielding a median growth gap of 5.8 percentage points. The middle half of metros had growth gaps from approximately 0.5 to 11.2 percentage points. These results indicate that home values grew faster than rents in much of the sample, though the size and direction of the difference varied considerably by location.

Table 1: Cumulative growth across the 100 metro areas, January 2020–August 2026

| Metric | Minimum | 25th pct. | Median | 75th pct. | Maximum |
|---|---|---|---|---|---|
| Home-value growth (%) | 5.9 | 40.8 | 49.5 | 54.5 | 79.9 |
| Rent growth (%) | 15.8 | 36.4 | 43.0 | 50.5 | 63.3 |
| Growth gap (percentage points) | -23.8 | 0.5 | 5.8 | 11.2 | 29.4 |
| PTR change | -3.1 | 0.0 | 0.7 | 1.4 | 3.4 |

The standard deviation of cumulative home-value growth was about 12.3 percentage points, compared with 10.3 points for rent growth. Thus, price growth varied more across metros than rent growth did in this sample. Approximately one quarter of metros experienced little increase or a decrease in PTR, as indicated by a 25th-percentile PTR change near zero.

## 5.2 The Price-to-Rent Ratio Peaked in Late 2022 and Then Eased

The median PTR ratio increased from approximately 16.7 in January 2020 to a peak near 18.1 in late 2022, then declined to about 17.2 by August 2026. The decline from the peak was approximately 0.9 ratio points, but the August 2026 median remained around 0.5 points above its January 2020 level. This suggests that the relative price of buying compared with the annualized rent measure rose during the earlier part of the period and later moderated without fully returning to its starting point.

Across all observed metro-months, the PTR ratio averaged about 18.6 and ranged from approximately 11.7 to 39.9. San Jose recorded the highest readings in the sample, with a peak of about 39.9 in April 2022. These values should be interpreted as relative price-to-rent indicators, not direct evidence that a specific home is overvalued. Mortgage costs, taxes, maintenance, expected tenure, and household circumstances are not included in the ratio.

## 5.3 The Gap Between Home Values and Rents Widened After 2020

When both series are indexed to 100 in January 2020, home values reached approximately 150 by August 2026, while rents reached about 143. This is consistent with the cumulative growth estimates: home values rose by about 48.1% on average, and rents rose by about 42.3%. The two series moved more closely together through late 2020, after which the gap widened during the 2021–2022 surge. Both series weakened from late 2022 into early 2023, and the gap later narrowed from a larger peak, although home values remained ahead of rents at the end of the period.

Year-over-year growth rates also cooled substantially from their pandemic-era highs. In 2021, home-value growth was approximately 15%, compared with rent growth of about 10%. Rent growth peaked around 13% in 2022, close to home-value growth of approximately 14%. By 2025 and 2026, both rates had slowed; the 2026 figures are partial-year observations, and the notebook's presentation indicates that rents were growing slightly faster than home values in the available 2026 observations. This recent reversal in growth rates is important, but it does not erase the cumulative difference built up earlier in the period.

## 5.4 Metro-Level Differences Were Substantial

The distribution of growth divergence shows that the national-level summary hides important local variation. McAllen, Texas, had the largest positive gap, at approximately 29 percentage points. Rochester and Syracuse, New York, followed with gaps of about 22 points. Three Connecticut metros also appeared among the ten largest positive gaps. The largest negative gap was approximately $-23.8$ percentage points, in a metro where home values rose only about 6% while rents increased by around 30%.

Austin, Texas, had the slowest rent growth in the sample, at approximately 15.8%, while home values increased by about 26.3%. The metro with the fastest cumulative home-value growth recorded an increase of nearly 80%, compared with rent growth of about 63%. The largest increases in PTR were in San Jose (about +3.4), Salt Lake City (about +3.0), and McAllen (about +2.7). McAllen, Salt Lake City, Nashville, and Charlotte appeared in both the top-ten growth-gap list and the top-ten PTR-increase list. These findings suggest that strong price growth relative to rent growth occurred in several different types of housing markets, rather than only in the most expensive coastal metros.

## 5.5 Rent Growth Was Less Informative About Prices One Year Later

The contemporaneous correlation between rent and home-value year-over-year growth was approximately 0.85. The correlation weakened as the lag increased, reaching around 0.41 at twelve months and approximately 0.13 at eighteen months, before a modest rebound to about 0.21 at twenty-four months. This pattern indicates that rent and price growth tended to move together at the same time, but their relationship was much weaker at longer horizons in this exploratory analysis.

The twelve-month-ahead regression produced an estimated slope of about 0.54 and an $R^2$ of approximately 0.17. In this model, one percentage point higher rent growth was associated with about 0.54 percentage points higher home-value growth twelve months later, on average. The $R^2$ indicates that rent growth alone accounted for only about 17% of the

variation in the regression's twelve-month-ahead home-value growth outcome. The remaining variation reflects factors not captured by this one-predictor model, as well as noise and model limitations. This result should not be interpreted as proof that rent growth causes later home-price growth.

### 5.6 Three Broad Growth Groups Emerged

The K-means analysis identified three descriptive groups based on cumulative home-value and rent growth:

Table 2: Average cumulative growth by K-means group

| Group | Home-value growth | Rent growth | Growth gap |
|---|---|---|---|
| Fast growers | 58.3% | 52.8% | 5.5 pp |
| Middle group | 47.8% | 40.3% | 7.5 pp |
| Slow growers | 30.6% | 28.1% | 2.5 pp |

The fast-growth group had the highest average cumulative growth in both measures. The slow-growth group had the lowest growth rates. The middle group had the widest average gap between home-value and rent growth, at about 7.5 percentage points. This shows that the largest difference between the two measures did not necessarily occur in the group with the fastest absolute growth. Since the clustering uses only two features and fixes the number of groups at three, it is best understood as a compact summary of the sample rather than a definitive typology of U.S. housing markets.

## 6 Stakeholders and Practical Implications

The results may be useful to several groups, with different decisions and caveats for each:

- **Prospective homebuyers and renters:** Metro-level trends can provide context for comparing how purchase values and rents have changed. However, individuals should also consider their income, down payment, mortgage rate, property taxes, insurance, maintenance, expected time in the home, and personal circumstances.
- **Policymakers and city planners:** Differences across metros can help identify places where purchase prices have grown much faster than rents or where rent growth has outpaced home values. These patterns may inform further analysis of housing supply, zoning, affordability programs, and local demand, but the project does not identify the causes of these differences.

- **Banks and mortgage lenders:** The analysis offers a high-level view of regional housing-market dynamics that may complement broader monitoring of mortgage demand and local market exposure. PTR alone is not a credit-risk measure and should not replace borrower-level underwriting or other risk indicators.
- **Investors and landlords:** The comparison can serve as a screening measure for further research into rental yields and relative valuations. It does not directly estimate returns because operating expenses, financing, taxes, vacancy, and future price or rent changes are not modeled.
- **Researchers and analysts:** The reproducible panel and feature definitions provide a basis for extensions incorporating income, mortgage rates, housing starts, inventory, migration, and other economic indicators.

# 7 Limitations

Several limitations affect the interpretation of these findings.

1. **Index data rather than household-level data.** ZHVI and ZORI summarize typical market conditions. They do not represent every home, lease, household, or transaction, and their underlying construction methods differ.
2. **Limited geographic coverage.** The analysis focuses on the 100 largest metros with sufficient overlapping data. Results may not generalize to smaller metros, rural areas, or every U.S. housing market.
3. **Uneven coverage and missing values.** ZORI has less complete historical coverage in some areas. Applying a minimum-coverage rule improves comparability but may exclude metros with different market characteristics. The final analysis should document the exact retained sample and row count generated by the notebook.
4. **Choice of start and end dates.** Cumulative growth compares endpoints and can hide substantial interim fluctuations. Results may differ if the period begins or ends in a different month. Observations for 2026 cover only part of the calendar year.
5. **PTR is not a full affordability measure.** The ratio omits mortgage interest rates, financing terms, taxes, insurance, maintenance, transaction costs, household income, and the expected duration of homeownership. A higher PTR does not by itself establish overvaluation or prove that renting is the better choice.

6. **Associational regression.** The lagged regression uses a simple specification and does not control for important confounders or account fully for panel dependence. Its low $R^2$ indicates limited explanatory power, and the estimated relationship is not causal.

7. **Clustering choices.** K-means results depend on the features, standardization, initial centroids, and the chosen value $k = 3$. Other choices could produce different groupings.

# 8 Conclusion

From January 2020 through August 2026, home values increased more than rents on average across the 100 selected U.S. metropolitan areas. Mean cumulative home-value growth was approximately 48.1%, compared with 42.3% rent growth, and the median growth gap was 5.8 percentage points. The median price-to-rent ratio rose from about 16.7 at the beginning of the period to a peak near 18.1 in late 2022, then eased to around 17.2 by August 2026. It therefore moderated after its peak but remained above its pre-surge level.

The cross-metro results reveal considerable heterogeneity. McAllen had the largest positive difference between home-value and rent growth, while San Jose experienced the largest increase in the price-to-rent ratio. K-means clustering summarized the sample into fast-, middle-, and slow-growth groups, with the middle group showing the largest average gap between home-value and rent growth. Finally, although rent and home-value growth were strongly correlated contemporaneously, rent growth alone had limited explanatory power for home-value growth twelve months later, with an $R^2$ of about 0.17.

Overall, the project demonstrates how public housing indices and reproducible data-analytics methods can characterize changing housing-market conditions across places and over time. The findings are most useful as a comparative starting point rather than a standalone affordability score, investment recommendation, or causal explanation. A natural next step is to incorporate mortgage rates, household incomes, housing supply, and other local economic variables to better understand why price-to-rent patterns differ across metros and how those differences relate to actual household affordability.

# Reproducibility and Data Availability

The analysis is implemented in the accompanying Jupyter notebook, `Mini_Project_Thao_Nguyen.ipynb`. The repository includes the downloaded ZHVI and ZORI CSV files in the `Data/` directory, subject to the applicable terms of the source data. To reproduce the analysis, run the notebook from the repository root and execute the cells in order. Because data files and code may be updated, the numerical results in this report should be checked against the final version of the notebook before submission.

# References

# References

[1] Zillow Group, Inc. (2026). *Zillow Research: Housing Data*. Zillow Home Value Index (ZHVI) and Zillow Observed Rent Index (ZORI). Available at: https://www.zillow.com/research/data/. Accessed October 2026.